Large Hotel Prototype Review

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On July 30, 2026, the Summary of Major Comments section was updated to follow the structure of four groups that later reviews contained. All changes reflect putting the comments into groups unless otherwise noted in square brackets [].

Large Hotel Scorecard Spreadsheet

Summary of Major Comments

The following comments represent errors found when comparing values from the scorecard to the source cited in the scorecard:

- Fix envelope (LHSC24-Envelope, LHSC25-Envelope)
- SHGC for non-north should be used from Title 24-2001 (LHSC28-Envelope)
- Infiltration value for 2008-2018, and 2019+, not consistent with reference (LHSC31-Envelope)
- Error in guest room occupant density in zone Room_5_FloorM (LHSC35-Zones, LHSC57-Ventilation)
- Boiler efficiency for 2008-2018 needs to be fixed (LHSC45-HVAC System)
- The water heater fuel type is wrong in three zones (LHSC60-Water Heater)
- Water heater set point not consistent with reference (LHSC63-Water Heater)
- Elevator load not consistent with ACM 2022 (LHSC77-Equipment)

The following are the highest priority comments:

- Number of floors too small (LHSC10-Prototype)
- The fraction of area that is guest rooms is too low (LHSC20-Prototype)
- Basement office zone as office is huge (LHSC21-Prototype, LHSC33-Zones, LHSC34-Zones, LHSC57-Ventilation)
- Guestroom occupancy is insufficiently diverse (LHSC36-Zones)
- Guestroom ventilation rate too low (LHSC57-Ventilation)
- Too many people assumed in the lobby (LHSC37-Zones, LHSC57-Ventilation)

- Guestroom thermostat settings are insufficiently diverse (LHSC56-Thermostat)
- Guest room lighting power density is high (LHSC69-Interior Lights)

The following comments represent other important issues found:

- Floor area too small (LHSC9-Prototype)
- No shading from entrance overhang (LHSC14-Prototype)
- Floor-to-ceiling height too low (LHSC16-Prototype)
- Fan motor and drive efficiency should vary by size (LHSC52-HVAC System)
- Water heater sizing not sensitive to water main temperature (LHSC64-Water Heater)
- Hotel dining schedules show little breakfast impact (LHSC82-Schedules)
- Lobby HVAC availability schedule should be 24x7 (LHSC84-Schedules)

In addition, the following comments describe the lack of disclosed assumptions that make verifying the values in the scorecard nearly impossible:

- Where is the distribution of hotel floor area by floor area bins from? (LHSC5-Notes) [added to this list]
- What is the basis of using non-operable windows (LHSC30-Envelope) [added to this list]
- Why is the dining on the top floor? (LHSC39-Zones) [added to this list]

Background

The review of the "score card," which is a spreadsheet summarizing the model assumptions, is broken down by spreadsheet tab, which includes:

- Notes
- Prototype
- Envelope
- Zones
- HVAC System
- Thermostat
- Ventilation
- Water Heater
- Ext Lighting
- Interior Lighting
- Equipment
- Schedules

- Weights
- PV
- EV Charger

The comments on these tabs are labeled using the following format LHSCx-<TAB>-<CELLRANGE>), where <TAB> is the name of the tab in the spreadsheet and <CELLRANGE> indicates the

LHSC1-Notes (n/a)

Sometimes, the specific benefits of code changes are not captured in a set of prototype models because they do not include a building with a specific feature. This has become an issue for PNNL, where a great fraction of the addenda for 90.1 with energy savings cannot be quantified using the models. See Table 4 of PNNL's "ANSI/ASHRAE/IES Standard 90.1-2022: Energy Savings Analysis - February 2024," where many addenda with energy savings were not included in the quantitative analysis and instead were just discussed qualitatively. 23 of the 39 were not captured quantitatively. In some ways, general prototypes cause this exclusion of understanding impacts, rather than custom prototypes that look at a particular measure. I'm not suggesting a change, but it is a warning that a good system of prototypes still does not answer all questions. Note this is the same comment as PPD1 in my single-family residential prototype review.

LHSC2-Notes (n/a)

The impact weight of a measure can be misleading when using a standardized set of prototype models. The scenario is when a measure is part of a technology that is represented in only one or a few prototype buildings in the set. In that case, the impact of the measure may be more indicative of the weight of that building rather than the weight of the technology in general. Take air-cooled chillers, for example. Perhaps they appear in only one prototype model for commercial buildings. An improvement measure for air-cooled chilled water plants could be significant, but the weight of one building may show that it has a small impact. This is because the weight of one building is based on the building type weighting, which may have nothing to do with the weighting of the air-cooled chiller market. In fact, it would be impossible to weigh the market size for every technology across a standardized set of building prototypes since the data usually does not exist. This subtlety makes it important to understand the frequency of appearance of a given technology in the set of prototypes, which is very important to disclose when assessing the overall impact of

a measure. Note this is the same comment as PPD2 in my single-family residential prototype review.

LHSC3-Notes (A1:C21)

Additional data sources could be added to the list shown in the spreadsheet:

- American Hotel and Lodging Association (AHLA) 2018 Lodging Survey https://www.ahla.com/sites/default/files/2018%20Lodging%20Survey%20Final.pdf
- HVS U.S. Hotel Development Cost Survey 2024 https://www.hvs.com/article/9999-HVS-US-Hotel-Development-Cost-Survey-2024
- Northwest Energy Efficiency Alliance – Commercial Building Stock Assessment 2019 https://neea.org/commercial-building-stock-assessment/
- ENERGY STAR Score for Hotels in the United States https://web.archive.org/web/20240513175940/https://www.energystar.gov/sites/default/files/tools/Hotel.pdf
- Commercial Buildings Energy Consumption Survey by DOE EIA https://www.eia.gov/consumption/commercial/
- Energy Efficiency Indicators for Hotel Buildings https://www.mdpi.com/2071-1050/13/4/1754
- Natural Resources Canada – Survey of Commercial and Institutional Energy Use 2019 https://oee.nrcan.gc.ca/corporate/statistics/neud/dpa/menus/scieu/2019/tables.cfm
- ASHRAE Advanced Energy Design Guide for Highway Lodging https://www.ashrae.org/technical-resources/aedgs/30-percent-aedg-free-download
- Technical Support Document – The Development of the Advanced Energy Design Guide for Highway Lodging Buildings – PNNL 17875 https://www.osti.gov/biblio/939043
- Technical Support Document – 50% Energy Savings Design Technology Packages for Highway Lodging Buildings – PNNL 18773 https://www.pnnl.gov/main/publications/external/technical_reports/pnnl-18773.pdf

By utilizing additional sources of information, the validity of assumptions can be strengthened. Some parameters may be California-specific, but comparing assumptions across multiple databases lends credibility to any assumptions made, even when a California-specific database ultimately may be the source of the information. Tables showing parameters across multiple databases, and then values chosen being highlighted, can be a powerful expression of when parameters are solidly determined by being similar in multiple data sets versus being based on just a single data set. Please also note that some of the above references may include citations of relevant reports and papers that could be reviewed as further sources.

LHSC4-Notes (A39:C39)

Please elaborate on the statements “Selected T24” and “is based on the construction trend within each vintage bin and almost follow the office.” It is not very clear what this means, especially the “follow the office” part of the statement.

LHSC5-Notes (A47:C89)

The data shown on the distribution of hotel floor area by floor area bins and number of stories, the combination of those, hotels above and below 60 Ksf, and the large hotel median and average, does not cite a source. Please add a citation for this data.

LHSC6-Notes (n/a)

The hospitality industry seems to categorize hotels into

- Luxury
- Upscale
- Midscale
- Economy
- Independent

Given that the room size, amenities, type of occupancy, and other features may be different for these different categories, it would be good for the prototype to be put into one of these categories. Another set of groupings that the hospitality industry uses is:

- Urban

- Suburban
- Airport
- Interstate
- Resort
- Small Metro/Town

And the prototype should probably be described as one of these categories. In addition, if the properties that would impact energy use vary greatly across these groups, additional prototype variations may be warranted.

LHSC7-Notes (n/a)

One type of categorization that I frequently see is hotels with or without convention facilities. I would think this would have a major energy impact to keep many large rooms, each with high ceilings, conditioned when the design occupancy for them is very high density (when used with auditorium-style seating), and can have a range of occupancy profiles.

LHSC8-Notes (A75:B89)

The use of NUMPRO=1 is not explained. I think it means “Number of Projects,” but that is not explained at all. The phrase “all extended” is also used without explanation.

LHSC9-Prototype (A8:C8)

The chosen floor area of 122,115 sf for the prototype seems small based on the values shown on the Notes tab (B82:B89) of 149,000 sf or 153,000 sf.

LHSC10-Prototype (A10:C10)

The number of floors for the prototype of six seems small based on the weighted average value of 8.8 or 9.2 shown on the Notes tab (B82:B89).

LHSC11-Prototype (A7:C7)

Based on the window layout and shape, the top floor and the two bottom floors of the prototype must have some different functions from the other floors. Please provide an overview of space types on each group of floors.

LHSC12-Prototype (A12:C12, A41:C41)

The PNNL window-to-wall ratio of 30% seems high. Please confirm this number with other sources.

LHSC13-Prototype (A16:C16)

The glazing sill height for the ground floor is shown as 6 in. This seems too low for many of the spaces on the ground floor of a hotel, such as office spaces, recreational spaces, and guest rooms. It is appropriate for lobby windows. The 3.6 ft glazing sill height for other floors that are probably mostly guest rooms seems a little too high for a large hotel.

LHSC14-Prototype (A13:C13)

The model shows no shading geometry; this is very unlikely for the lobby entrance, which almost always has a large overhang over the entrance to protect visitors from rain when checking in or checking out of the hotel. Generally, the larger the hotel, the larger this protected area is, and often covers much of the large windows usually associated with a large hotel lobby.

LHSC15-Prototype (A7:C7)

The orientation of the building with the front of the building facing south will result in more high cooling load hours each day, but probably not as high peak loads if the building were oriented so that the lobby faced west. In that orientation, the solar loads on the large amount of window area would increase peak loads during morning and evening hours. Unless the prototype is rotated randomly, the orientation of the building with this aspect ratio is always going to impact loads. I would suggest a side study be conducted in a few climate zones, looking at the energy and peak demand impacts of different orientations.

LHSC16-Prototype (A14:C15)

The floor-to-ceiling height for the ground floor of 13 ft seems rather low. Most hotels that have the number of stories that this prototype has would have high ceilings at least in the lobby and other public areas on the ground floor where the entrance is. Also, the floor-to-floor and floor-to-ceiling heights are shown as the same, which is very unlikely for a building with VAV not to have significant plenum space. In fact, most hotels have a lot of plumbing and access to plenums since they are often designed with the understanding that rooms may be rearranged in the future, especially on floors with any common spaces or conference rooms.

LHSC17-Prototype (A30:Q61 – note these rows are hidden and are intended to be deleted prior to publication**)**

Some information in this section of the spreadsheet should not be deleted prior to publication. The portions related to the prototype being used (presumably the PNNL column) are not shown in other locations and should be retained.

LHSC18-Prototype (A49:C51, G35:Q50 – note these rows are hidden and are intended to be deleted prior to publication**)**

Values are shown in SI units with square meters and should be expressed as square feet to be consistent with the IP units used on the rest of the spreadsheet.

LHSC19-Prototype (A49:C51, G36:Q37 – note these rows are hidden and are intended to be deleted prior to publication**)**

The use of a small and a large guest room is shown in the spreadsheet, but without any explanation. Some discussion of this and how they were apportioned in the building would make sense. Is this the difference between a suite and a normal room, or is this just the difference between two different style rooms?

LHSC20-Prototype (G36:L37 – note these rows are hidden and are intended to be deleted prior to publication**)**

The fraction of the total building area that comprises guest rooms in the CPUC model seems more reasonable at 64% than the PNNL prototype, which is only 46%. I would

encourage trying to find references to substantiate this low fraction of the total building area being guest rooms.

LHSC21-Prototype (G45:L45 – note these rows are hidden and are intended to be deleted prior to publication**) and Zones (A3:I25)**

The 19% (or shown as 17.4% on the Zones tab) fraction of the building that is shown as basement seems unrealistically high. If a hotel has that much basement area, it is probably being used as the laundry, storage, mechanical, refrigerated storage, and maybe even the kitchen. It is also common for the basement areas of large hotels to be used as conference rooms. If the basement is being used for conference rooms, the basement would need to have a higher ceiling than 8 ft (see A14:B15). I think it is very unlikely that a hotel would be developed with such a large basement that was not subdivided into useful spaces that would make the hotel more profitable.

LHSC22-Prototype (n/a)

The number of guest rooms is a primary metric used for hotels and should be described on this tab. Based on a sum of multipliers for guest rooms on the Zones tab, I think it is 179.

LHSC23-Envelope (A1:X37)

Individual rows were checked on this tab to check that they match the source:

External Wall Guestroom

- Row 3 - pre-1978 from ASHRAE 90.1-20022 Table A3.3.3.1 R-O - ok
- Row 4 – 1978-1997 – one error found, see LHSC24-Envelope – fix needed
- Row 5 – 1998-2007 from T24-2001 ok
- Row 6 – 2008-2018 from T24-2013 ok
- Row 7 – 2019+ from T24-2025 ok

External Wall NonRes

- Row 8 – Ext wall nonres pre-1978 from ASHRAE 90.1-20022 Table A3.3.3.1 R-O - ok
- Row 9 – 1978-1997 – one error found, see LHSC24-Envelope – fix needed
- Row 10 – 1998-2007 from T24-2001 ok
- Row 11 – 2008-2018 from T24-2013 ok

- Row 12 – 2019+ from T24-2025 ok

Roof

- Row 13- pre 1978 from T24-1992 ok (although see related comment LHSC26-Envelope)
- Row 14 -1978-1997 from T24-1992 ok
- Row 15 – 1998-2007 from T24-2001 ok
- Row 16 – 2008-2018 from T24-2013 ok
- Row 17 – 2019+ from T24-2025, two errors found, see LHSC25-Envelope – fix needed

Windows Guest Rooms

- Row 18 U-Factor – pre 1978 from average from HOF/T24-2013 – ok (although see related comment LHSC29-Envelope)
- Row 19 U-Factor - 1978-1997 from average from HOF/T24-2013 – ok (although see related comment LHSC29-Envelope)
- Row 20 U-Factor 1998-2007 based on T24-2001 ok
- Row 21 U-Factor – 2008-2018 from T24-2013 ok
- Row 22 U-Factor – 2019+ from T24-2025 ok
- Row 23 SHGC – pre 1978 from average from HOF/T24-2013 – ok (although see related comment LHSC29-Envelope)
- Row 24 SHGC - 1978-1997 from average from HOF/T24-2013 – ok (although see related comment LHSC29-Envelope)
- Row 25 SHGC 1998-2007 based on T24-2001 ok (although see related comment LHSC28-Envelope)
- Row 26 SHGC 2008-2018 from T24-2013 ok
- Row 27 SHGC – 2019+ from T24-2025 ok

Windows Non-Res

- Row 28 U-Factor – pre 1978 from average from HOF/T24-2013 – ok (although see related comment LHSC29-Envelope)
- Row 29 U-Factor - 1978-1997 from average from HOF/T24-2013 – ok (although see related comment LHSC29-Envelope)
- Row 30 U-Factor 1998-2007 based on T24-2001 ok
- Row 31 U-Factor – 2008-2018 from T24-2013 ok
- Row 32 U-Factor – 2019+ from T24-2025 ok

- Row 33 SHGC – pre 1978 from average from HOF/T24-2013 – ok (although see related comment LHSC29-Envelope)
- Row 34 SHGC - 1978-1997 from average from HOF/T24-2013 – ok (although see related comment LHSC29-Envelope)
- Row 35 SHGC 1998-2007 based on T24-2001 ok (although see related comment LHSCx-Envelope)
- Row 36 SHGC 2008-2018 from T24-2013 ok
- Row 37 SHGC – 2019+ from T24-2025 ok

LHSC24-Envelope (M4, M9)

The exterior wall guest room and nonresidential for 1978-1997 vintage is shown as 0.256, which is based on Rt=3.9, but the value from Title 24-1986 shows Rt=4.6, which should be a U-value of 0.217. This column value was probably copied from the CZ6 value by accident.

LHSC25-Envelope (K17, M17)

The roof U-Value for 2019+ is based on T24-2025, and the note says, “The lowest U-factor between Nonres and Guest room was considered,” but the values for CZ3 and CZ5 are incorrect. The U-Value shown for both CZ3 and CZ5 is 0.034, but the Nonres value from Table 140.3-B for “wood framed and other” shows 0.28 for both CZ3 and CZ5, which is lower. The 0.34 value is probably from Table 140.3-C for a hotel.

LHSC26-Envelope (A13:X14)

The roof values are based on Title 24-1992, as explained in the note: “Considering 30 years for roof/ insulation replacement. (Guest room requirement)” and the values are correct, but the presumption that T24-1992 is a good stand-in for this in 2025 seems incorrect. If one assumes that all these roofs were replaced 30 years ago, then Title 24-1995 would make more sense. Since the odds of all the roofs being replaced in 30 years are slight, a better assumption would be:

- The average value of roof requirements from Title 24 from 1995, 1998, 2001, 2005, 2008, 2010, 2013, 2016, 2019, 2022, or
- The value of roof requirements from 15 years ago, so using Title 24-2010 (less accurate but still justifiable).

LHSC27-Envelope (all)

Instead of using a single version of Title 24 to represent a range, perhaps it would make sense to use an average value for all versions of Title 24 within the time period. In other words, for the 1978-1997 range, values from the 1978, 1980, 1982, 1984, 1986, 1987, 1988, 1992, and 1995 versions of Title 24 should all be extracted and averaged together. Currently, from 1978-1997, uses values from Title 24-1992 only. For the time range 1998-2007, which could average the values from the Title 24 versions from 1998, 2001, and 2005 instead of using just the 2001 values. For the time range 2008-2018, we would average the values from Title 24 versions from 2008, 2010, 2013, and 2016 instead of using just the 2013 values.

LHSC28-Envelope (A25:X25, A35:X35)

For the SHGC for 1998-2007, the Title 24-2001 version was used, but that version has different SHGC values for north and non-north directions; however, the values from the table are for non-north only. The prototype for this version of the standard should probably use the north values for all fenestration facing north.

LHSC29-Envelope (A18:X19, A23:X24, A28:X29, A33:X34)

Averaging the U-Factor and SHGC between single and double-pane options leads to non-physically possible windows. The weighting between single and double pane is based on a statistic that showed 43% of large hotels that were built before 2000 were single pane from CBECS 2018. The weighted value is shown correctly, but the concept of ending up with a type of fenestration that is not physically possible may cause problems with simulation engines such as EnergyPlus that expect a window to be built up from a specific pane of glass with specific properties. It might be better to have a mixture of single-pane and double-pane windows in a model (although that is a different compromise with reality).

LHSC30-Envelope (many)

Assuming only fixed windows and no operable windows may be appropriate for high-rise hotels in some climates, but operable windows are common in many hotels in temperate climates.

LHSC31-Envelope (E39:X45)

The infiltration I_75 values of 1.1 for Pre 1978, 1978-1997, and 1998-2007 are ok, but the 2008-2018 and 2019+ values of 1.0 and 0.6, respectively, do not appear in the referenced location of Table 140.3-B. That table does indicate that some climate zones are required to have an air barrier, which should correspond to a value from the CASE report of 0.7. But instead, a value of 1.0 or 0.6 is shown for all zones. It is not clear how the values in the scorecard were determined.

Note there is a typo in the header, and it should be “6-sided”

This comment is identical to a comment in the warehouse review.

LHSC32-Envelope (E46)

The document that appears to be linked, titled “Infiltration_rate_report_14May2025-Short-Version.docx,” when clicked on, shows that the address is invalid. It is possible that this document would explain the information assumptions from the previous comment.

This comment is identical to a comment in the warehouse review.

LHSC33-Zones (D3)

This is the same comment as LHSCx-Prototype (G45:L45)

The 17.4% (or shown as 19% on the Prototype tab hidden rows) fraction of the building that is shown as basement seems unrealistically high. If a hotel has that much basement area, it is probably being used as the laundry, storage, mechanical, refrigerated storage, and maybe even the kitchen. It is also common for the basement areas of large hotels to be used as conference rooms. If the basement is being used for conference rooms, the basement would need to have a higher ceiling than 8 ft (see A14:B15). I think it is very unlikely that a hotel would be developed with such a large basement that was not subdivided into useful spaces that would make the hotel more profitable.

LHSC34-Zones (A3:I3)

Continuing from the previous comment concerning the basement, the basement is shown with a Space Type of office, and due to that 5 people/1000 ft2 occupant density, it ends up

containing 106 people. Not sure what they are all doing in the basement, but it probably isn't legal. No hotel of this size would need an office with that many people, maybe 5 people.

LHSC35-Zones (I15)

For the guest room in zone Room_5_FloorM, the people density is shown as 33 persons/1000 ft2, which is 10 times the density for this guest room as any other guest room, which uses 3 persons/1000 ft2. This may be a copy/paste error since 33 is used for other zones.

LHSC36-Zones (A11:I16, A18:I20)

For guest rooms, the number of occupants is shown as 1 for guest rooms (except for Room_5_FloorM, see previous comment), but is actually either 1.3 for the larger guest rooms or 0.8 for the smaller guest rooms based on column I "People" expressed in Persons/1000 ft2. Given the large fraction of the building that guest rooms represent, it is important to model them accurately. It might make more sense to try to derive an estimate of occupancy, based on literature suggested in LHSCx-Notes (A1:C21) or from other sources, to find how often a room has no-one, one person, a couple, three, or four people occupy hotel rooms and create a schedule that mixes up those integer values of people so that the overall occupancy matches the statistics. The use of lighting, equipment and water in rooms with zero to will clearly be different than one or four people.

LHSC37-Zones (H10)

The number of people in the lobby, 465, seems very large. For a hotel that has 179 rooms, that would mean every guest in the hotel would be there at once, along with one or two friends. This seems very unlikely, especially since the schedule (assembly/occupancy) would have 80% of this occupancy on every weekday from noon to 6 pm. In my experience, hotel lobbies for this size hotel are usually sparsely occupied except perhaps for a few evening hours.

LHSC38-Zones (general)

Please include the plan view layout of each group of floors on the tab instead of just one, and also identify the zone types, perhaps by color.

LHSC39-Zones (A9:I0, A22:I23)

The location of the dining room on the top floor, with the kitchen, and the location of the café/dining room on the first floor, seem inconsistent with the usual positioning of a kitchen. In a high-rise hotel, sometimes you see dining rooms on the top floor, but in this size hotel, I would expect that a dining room (presumably a restaurant) would be located with as easy access to the lobby and exterior entrances as possible, and that would be the first floor. This would encourage patrons from both hotel occupants and the local community. By locating both dining facilities on the first floor, the kitchen should also be on the first floor, which would provide direct access to both.

LHSC40-Zones (A21:I21)

The Hotel Function Area in a space called “Banquet_FloorT” is on the top floor and is more likely in this mid-size hotel to be in a basement or lobby level. It is also likely to have a taller ceiling than 10 ft.

LHSC41-Zones (x:x)

Mechanical room, Storage, and Commercial Laundry are more likely to be located in the basement rather than the first floor for a mid-size hotel.

LHSC42-HVAC System (A2:G6)

The overall HVAC system descriptions for existing and new construction seem reasonable.

LHSC43-HVAC System (E4)

The use of a single-speed fan for guest room FPFC for the existing prototype may be reasonable for older vintages, but for 2008-2018, it may be more appropriate to assume a multi-speed fan. They are quieter at low speeds, and the difference in cost for multi-speed fans (or ECM) has gone down so much that they may have been specified based on noise alone.

LHSC44-HVAC System (B12)

The year described as “Upgrade/Installed Year” should also include that this is the year of the version of Title 24 that is assumed.

LHSC45-HVAC System (A11:H17)

Chiller efficiencies

- Pre 1978, 1978-1997, 1998-2007 with Title 24-2008 ok
- 2008-2018 with Title 24-2013 ok
- New construction Title 24-2025 ok

Boiler efficiencies

- Pre 1978, 1978-1997, 1998-2007 with Title 24-2008 80% Et cannot verify (see LHSC46-HVAC System)
- 2008-2018 with Title 24-2013 – 84% need a fix - it should be 80%
- New construction Title 24-2025 – 90% ok
- New construction Title 24-2025 – 84% cannot verify (see LHSC47-HVAC System)

LHSC46-HVAC System (E13:H15)

The referenced table 110.2-K does not exist, and no table seems to exist for boiler efficiencies. I cannot verify the 80% Et.

LHSC47-HVAC System (E17:H17)

Neither of the referenced tables 110.2 J nor 110.2-D shows boiler efficiencies, so I cannot verify the 84% used. Is there a boiler efficiency table?

LHSC48-HVAC System (E16:F16)

The boiler efficiency table in Title 24-2013 is Table 110.2-K, not Table 110.2-J as shown.

LHSC49-HVAC System (E13:415)

The chiller efficiency table in Title 24-208 is Table 112-D, not Table 110.2-D as shown.

LHSC50-HVAC System (E13:415)

Confirming fan power inputs with PNL 20405 report

- Existing VAV TSP 5.58 – ok
- Existing FPFC TSP 1.33 – ok

Confirming fan power inputs with ACM 2025

- Existing VAV Fan Efficiency 0.65- ok
- Existing FPFC Fan Efficiency 0.65 – ok
- Existing fan motor and drive efficiency ~0.9 – ok with explanation shown
- New construction fan efficiency 0.65 - ok
- New construction VAV system W/cfm 0.947 – ok
- New construction FPFC 0.802 – ok
- New construction FPFC 0.35 - cannot verify (see LHSC53-HVAC System)
- New construction DOAS Supply Fan 0.626 - cannot verify (see LHSC53-HVAC System)
- New construction DOAS Exhaust Fan 0.144 - cannot verify (see LHSC53-HVAC System)
- New construction fan motor and drive efficiency – needs addressing (see comment LHSC52-HVAC System)

LHSC51-HVAC System (D23:D31)

The header indicates that the source for the Total Static Pressure is PNNL-20405, but that is only used for the existing vintage rows. The other rows are based on calculations using the fan formula and inputs from the ACM. Please correct the header.

LHSC52-HVAC System (F28:F31)

For new construction, a fixed value of 0.9 for the fan motor and drive efficiency is not correct, and instead, the size of the motor and ACM 2025 Table 13 should be used. This is described as the process for existing vintages but not for new construction.

LHSC53-HVAC System (G29:G31)

Several W/cfm values from the fan power table cannot be verified and cannot be found in the ACM 2025 source as described:

- New construction FPFC 0.35
- New construction DOAS Supply Fan 0.626
- New construction DOAS Exhaust Fan 0.144

Please add a description in the table of how these values were derived.

LHSC54-Thermostat (A7:R10)

The corridor heating (68°F constant) and cooling (78°F constant) setpoints for corridors are based on the Appendix 5.4B spreadsheet (the version I referenced was named "CEC-400-2022-009-CMF-REV2-AP5.4B.xlsx") and the "ResidentialCommon" tab. Please add the tab name to the reference.

LHSC55-Thermostat (A3:R6)

The non-res common space heating (70°F with 60°F setback) and cooling (75°F with 85°F) setpoints seem to be based on the Appendix 5.4B spreadsheet (the version I referenced was named "CEC-400-2022-009-CMF-REV2-AP5.4B.xlsx") and the "Assembly" tab. Please add the tab name to the reference.

LHSC56-Thermostat (A11:R14)

For guestrooms, the thermostat setting is a complicated topic. Using a 70°F constant setpoint for heating and 72°F based on a default is a simple approach, but I would recommend a more comprehensive approach for the prototype. Some considerations to include would be:

- Diversity of temperatures that hotel guests find comfortable
- Hotel guests do not pay extra for energy use to obtain greater comfort
- Oversetting of thermostats upon initial entry of the room (exaggeration of setpoint need)
- Thermostats with occupancy sensors and their impact on sleeping guests
- BAS resets setpoints when no guest is present

- Hotel staff resetting setpoints

To try to capture these effects, use several different schedules for guest rooms that have a wide diversity of values, including deep setbacks for unoccupied rooms.

LHSC57-Ventilation (B2:B24)

The number of occupants in different zones directly impacts the outdoor airflow rates, and comments on the number of occupants were already covered in comments: LHSC34-Zones, LHSC35-Zones, LHSC36-Zones, LHSC37-Zones. Specifically, the guest room ventilation needs to be based on maximum expected occupancy, which is generally four (since hotels often switch rooms that can have two double or queen beds with rooms with a single queen or king bed).

LHSC58-Ventilation (A1:K24)

No differentiation by vintage was included for outdoor airflow. Earlier vintages of hotels probably had significantly different ventilation requirements. Since this often impacts the sizing of ducts, intake vents, and air-handling equipment, it is unlikely to be upgraded very often.

LHSC59-Ventilation (D3:F24, G3:J24, A29:F31)

The cfm/person, cfm/ft2, and required exhaust rate columns were confirmed with the source tables in Title 24-2025. The formulas for determining the total outdoor air are consistent with Title 24-2025. The exhaust hood assumptions are ok.

LHSC60-Water Heater (B11:B32, E11:E32)

The peak water heating consumption table columns for “Baseline WH Type” and “Hot Water Usage Specification” were verified against the source T24, 2022, Appendix 5.4A, sheet: “2022 SpaceFuncData-Input”, and all were ok except the Baseline WH Types for Lobby (expected electric), Room_1_FloorM (expected gas), Room_2_FloorM (expected gas).

LHSC61-Water Heater (A4:B5, A11:B32)

The use of both electric and gas water heating equipment for new construction in the same building is quite unusual, except in the case of small electric water heaters being used under remote washrooms for sinks. This is unlikely in a hotel that has extensive piping to serve all the guest rooms already. Based on this, all water heating should probably be gas.

LHSC62-Water Heater (A3:C5)

The efficiency of the gas water heaters for both existing and new construction may not be the best assumption. Older water heaters may have been less efficient initially or may decrease in efficiency due to corrosion of the heat exchanger or burner.

LHSC63-Water Heater (A44:B44)

The design supply temperature is shown as 135°F, but the source for this Appendix 5.4B (called CEC-400-2022-009-CMF-REV2-AP5.4B.xlsx) on the "ResidentialLiving" tab shows a constant 130°F as the "WtrHtrSetpt."

LHSC64-Water Heater (A45:B45)

The cold water supply temperature of 55°F shown in the scorecard is consistent with the citation of the ACM 5.9.2 Rated Capacity, but better data is available that may be closer to how water heaters are actually sized using the local water mains temperature. The Appendix 5.4B (called CEC-400-2022-009-CMF-REV2-AP5.4B.xlsx) includes a WaterMain tab and WaterMainMonth tab that show climate-specific values for the entering water temperature. The lowest of these values for each climate zone is more likely to be used for sizing water heaters; the monthly values vary from 41.4 in CZ16 to 63.2 in CZ15. Using these values would have a large impact on the sizing of the water heater.

LHSC65-Water Heater (A46:B47, G40)

The water heater system factor of 0.6 and reheat multiplier of 0.4 are shown as described in the ACM reference. The standby loss equation looks correct for gas storage water heaters. The standby loss percentage equation for electric water heaters could not be verified.

LHSC66-Interior Lights (D14)

The reference to ASHRAE 90.1-2022 for guest rooms includes “PNNL” unnecessarily since 90.1 is an ASHRAE publication.

LHSC67-Interior Lights (B4:B25)

LPDs are consistent with the cited source for new construction.

LHSC68-Interior Lights (E4:E11, E18, E22:E25)

LPDs are consistent with the cited source for existing buildings, except that the LPD for the guest room value of 1.27 was not checked.

LHSC69-Interior Lights (E12:E17, E19:E21)

The lighting power density for the guest room, with a value of 1.27, seems very high. A value of 0.41 is suggested in the notes and would be consistent with the 90.1-2019 version value.

LHSC70-Interior Lights (E4:G25, A27)

The use of T24-2019 for existing buildings based on the comment:

“Hotels are expected to update their lighting systems more frequently than offices. Therefore, the code edition of 2019 was considered as the reference for LPD of the existing prototypes.”

This seems a bit optimistic for how often a hotel does a lighting refresh. Using Title 24-2013 might be more realistic. Of course, the same type of comment applies as stated for other system-related replacements. It might be better to average several different versions of Title 24 together rather than assume that all existing buildings have updated their systems during a specific time. In this case, I would suggest that versions of Title 24 from 2008, 2010, 2013, 2016, and 2019 all be examined, and an engineering estimate of the weight of each be used to determine the LPD for each space.

LHSC71-Ext Lights (E9:E15)

The lighting power allowances from Title 24-2025 were checked and are ok

LHSC72-Ext Lights (n/a)

Where are the exterior lighting assumptions for the vintage models? One assumes that those should be higher.

LHSC73-Ext Lights (B9:C15)

Please document the sources used for assumptions for computed areas and counts. It mentions the 2019 Outdoor Lighting Case Report and PNNL 90.1-2010 TSD, but those are both big, complicated documents. More specific references would enable the values to be checked.

LHSC74-Equipment (E3:E5, E9:E16, E18:E20)

The reference "Plug load and schedules project with CEC, 2025" could not be found to verify many of the numbers shown on the table.

LHSC75-Equipment (C2:D24)

The Plug Load and Gas Power Density for all numbers using Title 24-2022 Appendix 5.4A were all verified using the spreadsheet titled "CEC-400-2022-009-CMF-REV2-AP5.4A_0.xlsx" on the "2022 SpaceFuncData-Input" tab.

Numbers based on "Plug load and schedules project with CEC, 2025" were not verified (see previous comment LHSC74-Equipment).

LHSC76-Equipment (A23:F36)

The wrong factor was used in the article cited in the scorecard for the number of elevators. In the article, office buildings, hotels, residential buildings, and medical buildings were each provided with a rule of thumb for the number of elevators. The scorecard uses the rule of "one elevator for approximately 50,000 ft^2 in use" and "for every two floors or two and a half floors, consider adding one more elevator," but that is the rule for office buildings. Instead, the rule for hotels is "provide one elevator for every 75 rooms or so, with a minimum of one elevator up to three floors and add one more for four floors." Using the office building rule of thumb results in six elevators, but using the hotel rule results in only

four elevators. It also states an adder of “to help with room service and maid service, add a service elevator for every two passenger elevators,” and when this is factored in, the result is six elevators. So, the quantity is the same as what was assumed, but the justification should be updated in the scorecard. In addition, the schedule of use for the block of guest elevators and service elevators should be different.

LHSC77-Equipment (C29, A37:F38)

The elevator load suggestion from ACM 2022 of 10 kW per elevator (I could not find a specific reference to this value), resulting in 60,000 W, was not used in the elevator design load value in cell C29, and instead, the PNNL value of 75,759 W was used.

LHSC78-Equipment (A27:D29)

Using the same elevator design load for new construction and existing vintages seems incorrect. Elevators, while serviced regularly, are not typically updated that often, and existing buildings could easily have elevators that are less efficient. A major upgrade in elevator requirements has happened in 90.1 in one of the last few cycles, and I expect the same to be occurring in Title 24.

LHSC79-Schedules (all)

No checking of schedule values with the original source was performed.

LHSC80-Schedules (A7:AI9)

The Hotel Banquet area (also referred to as the Hotel Function Area), having 80% occupancy from noon to 6 pm on weekdays, seems like very high occupancy to me. In my experience, mid-size, non-convention center hotels have facilities like this, but they are hardly ever occupied. Occasional birthday parties, graduation parties, and corporate events. I would expect that for most days they are unoccupied, and perhaps the schedule should just show a random day (one or two days per week) with 80% occupancy for two or three hours for each day.

LHSC81-Schedules (A4:AI6)

The elevator schedule used does not match what appears in ACM Appendix 5.4B on the ResidentialLiving tab.

LHSC82-Schedules (A7:AI9, A34:AI36)

For the hotel dining and café zones, the “assembly occupancy” is probably not an appropriate schedule since it shows 0% occupancy from 6 am to 9 am, and 20% occupancy from 9 am to 11 am during weekdays. These are often some of the busiest times for hotel dining facilities. Similarly, 80% occupancy from noon to 7 pm is probably not correct either, as a hotel dining room usually has low occupancy except for a bump at lunch time. Perhaps a different schedule would make sense.

If the dining occupancy schedule is revised, the gas equipment usage should be revised similarly.

LHSC83-Schedules (AC13:AI21)

The fourth, fifth, and sixth graphs are labeled “Assembly,” but the heading in column A does not agree.

LHSC84-Schedules (A25:AI27, A31:AI33)

The lobby HVAC availability schedule should probably be 24x7, and the infiltration for the lobby should match that. Depending on how open the retail and dining areas on the first floor are (they often share a large open space with the lobby), they may need to have the same schedule.

LHSC85-Schedules (A50:AI52)

There is no corridor service hot water usage according to the zone tab, so this schedule is not necessary.